

Numerical Optimization for Control

Course 062047 - M.Sc. Programme in Automation and Control Engineering

Suggestions to draft a high-quality technical document (report, thesis)

Text

- Plan before writing. Define the structure and the logical progression of the text you're going to write, thinking about the intended audience/purpose of the document;
- Write simple sentences. Consider breaking longer sentences into shorter ones;
- Use either first person plural, or third person singular. Try not to mix (with exceptions, such as the summary);
- Define each acronym at the first time of use. Consider including a list of acronym if needed to improve the clarity (for example, if there are many acronyms);
- Carefully check the text to correct possible typos or grammar errors. Double-check terms that you're not sure about, avoid "false friends";
- Read and improve your text many times, check each paragraph/section as well as the overall structure of the document and progression of its various parts. Try to read it with the eyes of someone who has general scientific/engineering knowledge but no knowledge of the specific topic of the report. Avoid repetitions, avoid using too early some concepts that are introduced only later on in the text;
- Keep a homogeneous format (fonts, bulleted lists, enumerated lists, captions etc).

Notation

- Use different symbols for different variables;
- Always use the same symbol for the same variable;
- Always define/explain symbols at the first time of use (ALL symbols, also the apparently trivial ones, such as the time variable);
- Consider including a list of symbols if needed, for example if there are many symbols;
- Define explicitly non-standard notation, either at the first time it's used, or in a preliminary notation section;
- Be precise in mathematical expressions (e.g. matrices in matrix operations must have the right dimensions).

Figures

- Aim for best quality/size ratio, for example use vector format for plots and technical diagrams (you can generate them as pdfs in vector format from Matlab and Powerpoint, and import them in Latex with pdfLatex), compressed images for pictures and color-dense plots or diagrams;
- Pay attention to the size, readability, and notation consistency of plot labels and titles;
- Write captions that highlight/explain important features of the image, try to strike a good balance between self-consistency and length of the captions.

Tables

- Keep a homogeneous style for the tables as well;
- Organize the tables so that they are not confusing, for example tables presenting the results of many tests. Highlight, for example in bold or by underlining, the results that are worth noticing inside the table, for example the best results when comparing different approaches.

Miscellaneous

- Scientific notation: try to be as consistent as possible when writing numbers (e.g., 0.01 vs. 10^{-2} , m/s vs. m s^{-1} etc.);
- Always include the unit of measure of physical quantities, try to use international units as much as possible;
- References: include complete references: check that year, page n., volume, type of source, authors, etc are all present. References must be available to your intended audience and ideally to anyone who reads the document;
- Use Latex to write the report.